

SubterraView Project Description Summary

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Overview

SubterraView is an augmented reality and artificial intelligence-based system designed to improve underground infrastructure management and worker safety. The platform compiles data from public utility records, city infrastructure databases, and crowdsourced field reports to generate a continuously updated underground utility map. Using an AI validation engine, the system cross-references incoming data, assigns confidence ratings, and resolves discrepancies to ensure high levels of reliability and accuracy. SubterraView is intended to serve construction workers, utility technicians, emergency responders, and public or private work organizations. By providing real-time underground visualization and hazard monitoring directly through smartphones, the system reduces reliance on the traditional 811 process, minimizes excavation delays, and helps prevent costly infrastructure damage and worker injuries.

Scopes of Work and Product

The scope of the work focuses on the digital detection, validation, visualization, and monitoring of underground infrastructure data within the broader environment of municipal utility management and safety coordination. Currently, underground infrastructure management relies on 811 notifications, fragmented utility databases, manual field verification, and surface paint markings. These processes are time-consuming and prone to data inconsistencies. Nearly 200,000 underground utility damage incidents occur annually, resulting in billions of dollars in losses and significant safety risks.

The scope of the product includes the development of a digital mapping and safety coordination platform that integrates AI-based record validation, augmented reality visualization of subsurface utilities, crowdsourced reporting capabilities, and real-time hazard alert notifications. Construction crews can visualize underground infrastructure directly through AR before and during excavation. Utility technicians can submit discrepancy reports or newly discovered utilities to improve map accuracy. Emergency responders can access a prioritized underground layout during incidents to determine safe entry points(view Figure 1).

Stakeholders + Constraints

SubterraView is for a wide range of clients, customers, and users ranging across many sectors(view Figure 1). The primary focus is on federal, public, and private construction groups who will be invested in the development and daily use of the app. While SubterraView is closely associated with the construction scene, it is designed to apply to everyday individuals to emergency responders, each bringing different levels of knowledge and technological experience. Behind the scenes, engineering and support teams will ensure the application remains functional and reliable through cases like model tuning updates and hardware support.

Beyond this, SubterraView will also be involved in ensuring to deal with several constraints. These include following the 811 federal mandate, maintaining accurate and reliable data through the AI engine, and ensuring compatibility across both iOS and Android platforms alongside any additional hardware SubterraView has such as hazards sensors.

Facts

- SubterraView depends on data provided by utility companies and municipal agencies.
- Hazard monitoring depends on sensor units detecting gas leaks, structural risks, or environmental abnormalities.
- The system supports construction planning, excavation safety, and emergency response coordination.

Assumptions

- Utility companies and municipal agencies will allow access to infrastructure data.
- Reliable internet connectivity will usually be available during system operation.
- AI validation will maintain approximately 90% infrastructure data accuracy.
- Infrastructure repair and regulatory compliance responsibilities remain with external organizations.

Figure 1

